

Mike Hall

Principal Software Engineer

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TECHNICAL LEADERSHIP & PRODUCTION SYSTEMS MODERNIZATION

Principal Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines 20+ years across software engineering and platform architecture with OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, and cloud infrastructure.

AREAS OF FOCUS

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|----|-------------------------------------|----|--------------------------------|
| 01 | Technical Leadership & Architecture | 02 | Legacy Systems & Modernization |
| 03 | Delivery & Observability | 04 | AI-Augmented Engineering |

EXPERIENCE

Development Manager — Founder Transition & Acquisition Handoff

EMR-Bear, a MedSuite company | Remote | May 2026 - August 2026

Engaged on a bounded 90-day interim assignment to conduct rapid architecture discovery, platform stabilization, and operational risk assessment for a multi-tenant healthcare SaaS platform serving 130+ clinics following a founder transition.

- Architecture Discovery & Operational Risk Assessment: Built a rapid 90-day system inventory and risk model spanning application codepaths, infrastructure dependencies, access boundaries, and operational risks.
- Production & Compliance Governance: Established release verification gates across the deployment pipeline of a platform servicing 130+ clinics, blocking unvalidated changes prior to HIPAA compliance review and business sign-off.
- Engineering Continuity: Created transparent component ownership, visible operational priorities, and collaborative decision rhythms for a distributed U.S.-Mexico engineering team.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Served as Senior Technical IC and Software Architecture Lead for the Acquisition lane of a regulated financial enterprise, combining hands-on legacy modernization with production stability, OpenTelemetry observability, and cross-team technical direction.

- **Technical Leadership & Team Consolidation:** Progressed from Digital Origination Team Lead to Software Architect for the Acquisition Lane, merging two attrition-impacted teams into a unified, high-functioning engineering unit before converting to Associate Director, Staff Engineer upon corporate architecture track restructuring.
- **ACQ Enablement Founding & Engineering Mandates:** Founded and technically led the ACQ Enablement team to protect core engineering concerns from product feature pressure: driving platform stabilization, legacy modernization, system process mapping, architecture documentation, PII remediation, and fraud investigations.
- **Observability & Enterprise Trace:** Founded and led the OpenTelemetry Working Group (OTel WG) to establish the Enterprise Trace across high-volume customer acquisition flows, connecting a complex Rails distributed monolith, MuleSoft, and IBM mainframe backends to eliminate blind spots and align engineering around real-time service maps.
- **State Machine Refactoring & Data Integrity:** Refactored fragile multi-step Rails workflow state machines, eliminating hidden lateral state mutations across asynchronous boundaries and resolving persistent data integrity failures in core acquisition flows.
- **Incident Response & Technical Escalation:** Served as final escalation point for high-severity outages, translating live root-cause investigations into stronger telemetry, explicit component ownership, and reusable system knowledge.
- **AI-Augmented Systems Investigation:** Introduced privacy-conscious local AI workflows for legacy code analysis and engineering support, expanding investigative capacity while keeping human review, security constraints, and source evidence explicit.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.

- **Search Infrastructure Simplification:** Replaced legacy external search clusters (Sphinx) with native MySQL full-text search and optimized relational indices, eliminating external service dependencies and reducing platform infrastructure footprint.
- **High-Volume Messaging Pipelines:** Designed and validated asynchronous batch delivery pipelines via SendGrid and Sidekiq, enforcing IP warmup protocols and deliverability safety for large-scale campaign operations.
- **Data Engine & Analytics:** Engineered complex relational analytical queries across multi-million row content catalogs, powering real-time data visualizations and international media publishing.
- **Zero-Downtime Platform Upgrades:** Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

SELECTED CURRENT WORK

Creator

Agent Tooling (MCP Servers & Agent Workflows) | Independent Project | 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- Bidirectional Knowledge Server: Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- Telemetry-Driven Diagnosis: Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- Compliance-Safe Observability: Telemetry reaches agents already PHI-scrubbed at the OpenTelemetry ingest edge, keeping a compliance-sensitive platform debuggable by an AI without exposing protected data.
- Local Inference Tooling: Built llama-mcp, exposing a local llama.cpp inference stack as callable tools for interrogation and validation, keeping routine inference on-box.
- Context as a Budgeted Resource: Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

Creator & Principal Architect

Phalanx Duel | Independent Project | 2022 - Present

Designing and building a real-time tactical game platform as a current hands-on laboratory for deterministic systems, product architecture, and controlled AI-assisted engineering.

- Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.
- Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.
- Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Creator & Forensic Engineer

Technical Conversation Archive | Independent Project | 2011 - Present

Recorded, and later restored, an archive of 214 technical interviews with practitioners from the Ruby, JVM, and Software Craftsmanship movements, using a local-only AI pipeline to transcribe, structure, and cross-link two decades of primary-source material into a searchable knowledge graph.

- Local-Only AI Restoration Pipeline: Built a Whisper and local-LLM pipeline that transcribes, diarizes, and structures long-form interview audio entirely on-device, keeping a large private media corpus off third-party inference services.
- Schema-Validated Content Platform: Runs on a contract-validated data layer: every interview, asset, and transcript is checked against an explicit schema at build time, with referential-integrity checks that fail the build rather than publishing broken data.
- Semantic Cross-Linking & Taxonomy Generation: Generates a topic taxonomy, knowledge graph, and semantic cross-links across the corpus, turning an unstructured media archive into navigable, individually indexed pages.
- Primary-Source Capture: Conducted the original interviews on-site at GOTO Conference, Software Craftsmanship North America, RailsConf, and WindyCityRails, with practitioners including Dave Thomas, Stuart Halloway, Corey Haines, Sandro Mancuso, and Micah Martin.
- Digital Archaeology: Recovered and reconstructed material from defunct platforms and web archives, reconciling incomplete metadata across sources to restore provenance for recordings that would otherwise have been lost.

CORE CAPABILITIES & TECHNOLOGIES

TECHNICAL LEADERSHIP & ARCHITECTURE

- Legacy Modernization & Architecture Discovery,
- Cross-Team Architecture & Technical Direction,
- Dependency Mapping & State Decoupling,
- Architecture Decision Records (ADRs) & RFCs

LEGACY SYSTEMS & MODERNIZATION

- Strangler-Fig Decoupling & API Contracts,
- Zero-Downtime Data Migration & Integrity,
- Incident Response & Root Cause Analysis,
- Distributed Systems & Asynchronous Processing

DELIVERY & OBSERVABILITY

- OpenTelemetry & Observability Architecture,
- Network Boundary & Access Control Auditing,
- Platform Enablement & Developer Velocity,
- Database Performance & SQL Analysis

AI-AUGMENTED ENGINEERING

- Model Context Protocol (MCP) Server Design,
- Agentic Workflows & Context-Isolated Subagents,
- LLM Orchestration & Deterministic Evaluation,
- Prompt Injection Defense & Input Guardrails,
- Vector Search & Semantic Retrieval (pgvector),
- Local-First Inference (llama.cpp, Ollama),
- AI Agent & Automation Interaction Mapping,
- Codebase Discovery & Domain Rule Extraction,
- Bounded, Human-Reviewed Agent Workflows

TECHNOLOGIES

- Ruby on Rails,
- PostgreSQL (including pgvector),
- Sidekiq & Redis,
- AWS & Cloud Infrastructure,
- Kubernetes, Docker & Linux,
- Claude Code, Anthropic & OpenAI APIs

SELECTED EARLIER EXPERIENCE

- **Senior Software Developer, BenchPrep** | March 2017 - February 2018 | Chicago, IL : Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote : Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Open-Source Transition Lead, Coderwall** | January 2014 - December 2014 | Remote : Hired as a contractor by founder Matt Deiters to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Engineering Learning & Development Business Partner, Groupon** | July 2011 - May 2013 | Chicago, IL : Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.

CAREER FOUNDATIONS

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.